

SchoolBell

Flutter App - Build Plan

Location-Aware Class Period Reminder for Android & Galaxy Watch 7

Developer	Alex
Target Device	Samsung Galaxy S25+ and Galaxy Watch 7
Framework	Flutter (Dart)
IDE	VS Code with Flutter extension
Platform	Android (APK sideload)
Est. Total Build Time	3 to 5 hours

1. App Overview

SchoolBell is a Flutter Android app that buzzes your phone and Galaxy Watch 7 3 minutes before each class period starts — but ONLY when you are physically at school. It uses GPS geofencing to detect your location, and only fires alerts Monday through Friday during school hours.

Core Features:

- Geofencing: define school location by GPS coordinates and radius (e.g. 150 meters)
- Time-based alerts: customizable list of period start times
- Smart filtering: only fires Mon-Fri, only when inside the geofence
- Vibration alert: buzz on phone, syncs to Galaxy Watch via Wear OS notification
- Simple UI: toggle on/off, view schedule, edit times
- Runs in background as a foreground service (survives screen off)

2. Tech Stack

Layer	Technology	Purpose
UI	Flutter + Dart	Cross-platform UI, settings screen
Geofencing	geolocator + geofence_service	Detect when user enters/exits school zone
Background	flutter_background_service	Keep app running when screen is off
Scheduling	flutter_local_notifications	Trigger timed vibration alerts
Storage	shared_preferences	Save period times and settings

Watch Sync	Wear OS notification bridge	Galaxy Watch receives phone notifications
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3. Flutter Dependencies (pubspec.yaml)

Add these packages to your pubspec.yaml:

```
dependencies: flutter: sdk: flutter geolocator: ^11.0.0 geofence_service: ^5.0.0
flutter_local_notifications: ^17.0.0 flutter_background_service: ^5.0.0
shared_preferences: ^2.0.0 permission_handler: ^11.0.0
```

4. Project File Structure

```
schoolbell/ lib/ main.dart # App entry point, background service init home_screen.dart
# Main UI: on/off toggle, schedule list settings_screen.dart # Edit period times, set
school location geofence_service.dart # Geofencing logic (enter/exit school)
scheduler.dart # Period time logic, Mon-Fri filter notification_service.dart #
Vibration + local notification trigger models/ period.dart # Period model (time, label)
android/ AndroidManifest.xml # Permissions: location, notifications, boot pubspec.yaml
```

5. Build Phases

Phase 1	Flutter Setup	20-30 min
Install Flutter SDK, configure VS Code extensions (Flutter + Dart), run flutter doctor, connect S25+ via USB for debug.		
Phase 2	Project Scaffold	15 min
Create new Flutter project, add dependencies to pubspec.yaml, set up Android permissions in AndroidManifest.xml (location, notifications, foreground service, boot receiver).		
Phase 3	Geofencing	45-60 min
Implement geofence_service to monitor school GPS coordinates. Store lat/lng in shared_preferences. Test enter/exit events with phone in hand.		
Phase 4	Scheduler Logic	30-45 min
Build period list model. Implement scheduler that checks current time against period list (minus 3 min), filters for Mon-Fri only, and only fires when inside geofence.		
Phase 5	Notifications	30 min
Set up flutter_local_notifications for vibration-only alerts. Configure background service so scheduler runs even when app is closed. Test on device.		
Phase 6	UI	30-45 min
Build HomeScreen (toggle, today's schedule) and SettingsScreen (add/edit/delete periods, set school location via map picker or manual coordinates).		
Phase 7	Watch Integration	15-20 min
Galaxy Watch 7 automatically mirrors phone notifications via Galaxy Wearable. Ensure notification channel is set to vibrate. Test buzz on watch.		

Phase 8 Build + Install APK

15 min

Run flutter build apk --release. Transfer APK to phone via USB or Google Drive. Enable Install unknown apps in Android settings and sideload.

6. Required Android Permissions

7. School GPS Coordinates

Archbishop Molloy High School, 83-53 Manton St, Briarwood, Queens, NY 11435

Latitude: 40.7074 Longitude: -73.8184 Geofence Radius: 150 meters (adjust based on testing)

8. Sample Period Schedule (Edit to Match Molloy)

Below is a placeholder schedule. You will enter the real Molloy bell times in the app settings. The app will buzz 3 minutes before each period start time.

Period	Start Time	Alert Fires At
Period 1	8:00 AM	7:57 AM
Period 2	8:50 AM	8:47 AM
Period 3	9:40 AM	9:37 AM
Period 4	10:30 AM	10:27 AM
Period 5	11:20 AM	11:17 AM
Period 6	12:10 PM	12:07 PM
Period 7	1:00 PM	12:57 PM
Period 8	1:50 PM	1:47 PM

9. Prompt to Use in VS Code with Claude

Import this PDF into Claude in VS Code (Claude extension), then paste the prompt below to begin coding:

I am building a Flutter Android app called SchoolBell. GOAL: Buzz my phone and Galaxy Watch 7 exactly 3 minutes before each class period starts, but ONLY when I am physically at school (geofence) and ONLY Monday through Friday. TECH STACK: - Flutter + Dart - geolocator + geofence_service (location) - flutter_local_notifications (alerts) - flutter_background_service (background) - shared_preferences (storage) SCHOOL LOCATION: - Archbishop Molloy HS, Briarwood, Queens, NY - Lat: 40.7074, Lng: -73.8184, Radius: 150m Please help me build this step by step, starting with Phase 1 (project setup and pubspec.yaml), then continuing through each phase as described in the attached build plan PDF.

